

PolyLens: A Recommender System for Groups of Users

Mark O'Connor, Dan Cosley, Joseph A. Konstan, and John Riedl
Department of Computer Science and Engineering, University of Minnesota,
Minneapolis, MN 55455 USA

{oconnor;cosley;konstan;riedl}@cs.umn.edu

Abstract. We present PolyLens, a new collaborative filtering recommender system designed to recommend items for groups of users, rather than for individuals. A group recommender is more appropriate and useful for domains in which several people participate in a single activity, as is often the case with movies and restaurants. We present an analysis of the primary design issues for group recommenders, including questions about the nature of groups, the rights of group members, social value functions for groups, and interfaces for displaying group recommendations. We then report on our PolyLens prototype and the lessons we learned from usage logs and surveys from a nine-month trial that included 819 users. We found that users not only valued group recommendations, but were willing to yield some privacy to get the benefits of group recommendations. Users valued an extension to the group recommender system that enabled them to invite non-members to participate, via email.

Introduction

Recommender systems (Resnick & Varian, 1997) help users faced with an overwhelming selection of items by identifying particular items that are likely to match each user's tastes or preferences (Schafer et al., 1999). The most sophisticated systems learn each user's tastes and provide personalized recommendations. Though several machine learning and personalization technologies can attempt to learn user preferences, automated collaborative filtering (Resnick et al., 1994; Shardanand & Maes, 1995) has become the

preferred real-time technology for personal recommendations, in part because it leverages the experiences of an entire community of users to provide high quality recommendations without detailed models of either content or user tastes.

To date, automated collaborative filtering systems have focused exclusively on recommending items to individuals. In some domains, such as Usenet News (Konstan et al., 1997; Resnick et al., 1994), this limitation is understandable. Few users read articles collectively. In other domains such as books or music (Shardanand & Maes, 1995), it is common both to enjoy the media alone and in groups. (Indeed, the MusicFX system (McCarthy & Anagnost, 1998), which did not attempt to use collaborative filtering, was designed specifically to address the challenge of selecting music for the often-large groups of people using a corporate gym.) Moreover certain items, among them restaurants, board games, and movies (Hill et al., 1995), are more commonly enjoyed in groups. Recommender systems that identify items such as movies for individuals do not address the user's key question, which is not "what movie should *I* see?" but rather "what movie should *we* see?"

This paper explores the design space of collaborative filtering recommenders for groups and presents our experience deploying the PolyLens group recommender to over 800 MovieLens users. The design space includes issues such as: What is the nature of a group? How are groups formed? How are recommendations computed for groups? What interfaces are best for sharing recommendations with groups? What are the privacy issues in showing recommendations to groups?

In our field trial with PolyLens we explored user experiences with one set of design choices. We kept detailed logs to measure how users formed groups, how they used the groups, and how the experience of users who used groups differed from the experience of users who did not use groups. We also surveyed the group users to learn their reactions to group recommendations, including their opinions about the value of the recommendations and the tradeoff in lost privacy.

In the next section, we present related work in recommender systems and in other systems with related group dynamics issues. We then introduce PolyLens and review the design space for group recommenders, looking at group properties and member rights, algorithms for group recommendation, and interfaces for displaying group recommendations. We follow with results of our user trial and survey, and we conclude with a discussion of lessons learned.

Related Work

Though we know of no previously published studies of groups in recommender systems, the work is related to previously published work on collaborative filtering, group formation, roles in collaborative systems, and awareness in collaborative systems.